AppliedMath

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New 3D Fractional-Order Zernike Moments for Shape Analysis

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Abstract

We introduce a new class of three-dimensional (3D) fractional-order Zernike moments by extending the radial component of classical 3D Zernike polynomials to admit fractional exponents. We derive explicit formulas for the fractional radial coefficients to establish orthogonality of the new Zernike functions in the unit ball and prove that they form a complete basis for square-integrable functions. We also present a three-way recurrence relation that enables efficient computation of high-order fractional moments. From these moments, we further construct rotationally invariant fractional-order Zernike descriptors for orientation-independent shape analysis. Numerical experiments demonstrate that fractional-order moments improve reconstruction accuracy and enhance discriminative performance compared with classical 3D Zernike moments. These results highlight the effectiveness of fractional Zernike functions as a robust basis for 3D shape representation and retrieval.

Keywords: shape retrieval; reconstruction; invariant descriptors; orthogonal polynomials; special functions; Zernike moments

1. Introduction

Image moments are defined as the integral of a basis function multiplied by the image function, which can also be seen as the inner product of these two functions. When the basis functions form an orthogonal set, the moments derived from them are called orthogonal moments. Orthogonality is crucial here, as it enables the moments to fully describe an image with minimal redundancy, with each moment representing a unique feature. As a result, reconstructing an image from a finite number of moments becomes possible. Despite this advantage, only higher-order moments carry the fine details of an image. The higher-order moments, however, are more sensitive to noise. Thus, constructing moments with higher accuracy has been a long-standing question.

Recently, fractional-order moments have been introduced to address these issues in two dimensions (2D). They are generally defined by replacing orthogonal polynomials with functions that allow fractional exponents. Some recent examples of work with fractional-order orthogonal moments in 2D include shifted Legendre moments [1], Fourier–Mellin moments [2], Zernike moments [3,4], Chebyshev moments [5], Jacobi–Fourier moments [6], Legendre–Fourier moments [7], shifted Gegenbauer moments [8], Laguerre moments [5,9], and Gaussian–Hermite moments [10]. Common to all of these works, fractional-order moments generalize the order of moments from integers to fractional numbers and maintain numerical stability as the order of moments increases.

The 2D version of fractional-order Zernike polynomials was introduced in [3] and applied to plant disease recognition in [11]. They are defined by transforming the radial

Check for updates

Received: 5 August 2026
Revised: 27 September 2026
Accepted: 9 October 2026
Published: 13 October 2026

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part of the traditional Zernike functions to include fractional orders. The moments derived from these functions were proved to be orthogonal on the unit disk, as their traditional counterparts. Because the change is made to the radial function but not to the angular part, their critical rotation invariance property is preserved. A quaternion form of 2D fractional-order Zernike moments was also introduced and used for image forgery detection [4]. While the 2D fractional-order Zernike moments have been established, the three-dimensional (3D) fractional-order Zernike moments are not available in the literature, to the best of our knowledge.

In this paper, we construct the new 3D fractional-order Zernike moments by introducing a new parameter in the radial part of the classical 3D Zernike polynomials [12,13] to allow for fractional exponents. We derive the coefficients in the radial function so that the new fractional Zernike functions continue to satisfy the orthogonality condition in the unit ball. We also verify that they form a complete basis for square-integrable functions. We further reformulate the fractional Zernike functions and provide a recurrence relation for efficient computation of the fractional-order moments. From these moments, we also compute rotationally invariant 3D fractional-order Zernike descriptors to compare shapes in different orientations and enable quick shape search in databases. We finally present numerical results, showing that extending the order of moments and descriptors to fractional numbers increases their accuracy in 3D shape retrieval.

2. 3D Fractional Zernike Functions

3D Zernike polynomials were introduced in [12], and the integer-order moments derived from them were first used in shape analysis in [13,14]. A fast algorithm for computing 3D Zernike moments was later given in [15]. They have been used in a diverse range of applications, with recent examples in [16–20]. In all of these methods, the functions used were polynomials; hence, the order of moments derived from them must be a natural number.

In this section, we define the 3D fractional Zernike functions by introducing a new parameter $\alpha > 0$ in the definition of the classical 3D Zernike polynomials [12]. We achieve this by modifying the radial functions in traditional polynomials. We ensure that the orthogonality of the new functions and the invariance of the descriptors derived from them are preserved. We first define the fractional radial functions by substituting r in the traditional radial polynomials by r^α for $\alpha > 0$ as follows:

$$R_{n,l;\alpha}(r) = r^{\alpha l} \sum_{\nu=0}^{k} q_{k,l;\alpha}^{\nu} \, r^{2\alpha\nu}, \tag{1}$$

where $n, l \in \mathbb{N}$, $0 \le l \le n$, $n-l$ is an even number, and $k = (n-l)/2$.

The 3D fractional Zernike functions will then be defined by:

$$Z_{n,l;\alpha}^{m}(r,\theta,\phi) = R_{n,l;\alpha}(r)\, Y_l^m(\theta,\phi), \tag{2}$$

where $0 \le l \le n$, $-l \le m \le l$, and $n-l$ is an even number. The spherical harmonic functions $Y_l^m(\theta,\phi)$ are complex-valued functions that form a complete orthogonal basis on the surface of the unit sphere S^2:

$$\frac{1}{4\pi} \int_0^{2\pi} \int_0^{\pi} Y_l^m(\theta,\phi)\, \overline{Y_{l'}^{m'}(\theta,\phi)}\, \sin(\theta)\, d\theta\, d\phi = \delta_{ll'}\delta^{mm'}. \tag{3}$$

To ensure that the new 3D fractional Zernike functions are orthogonal in the unit ball, we require:

$$3 \int_0^1 R_{n,l;\alpha}(r)\, R_{n',l;\alpha}(r)\, r^2\, dr = \delta_{nn'} \tag{4}$$

and determine a formula for the coefficients $q^{\nu}_{k,l;\alpha}$ in (1) satisfying this condition. Given n and n' and an l lower than both n and n', suppose $n-l$ and $n'-l$ are both even. If $k=\frac{n-l}{2}$ and $k'=\frac{n'-l}{2}$, we have $\delta_{nn'}=\delta_{kk'}$. Setting

$$R_{n',l;\alpha}(r)=r^{\alpha l}\sum_{\mu=0}^{k'}q^{\mu}_{k',l;\alpha}\,r^{2\alpha\nu},$$

the orthogonality condition in (4) yields:

$$3\sum_{\mu=0}^{k'}q^{\mu}_{k',l;\alpha}\sum_{\nu=0}^{k}q^{\nu}_{k,l;\alpha}\frac{1}{2\alpha(\nu+\mu+l)+3}=\delta_{kk'}. \tag{5}$$

Here, we can assume $k'\le k$ without loss of generality. Assuming all q-coefficients are nonzero and starting with $k'=0$ through $k'=k-1$, (5) yields

$$\sum_{\nu=0}^{k}q^{\nu}_{k,l;\alpha}\frac{1}{2\alpha(\nu+\mu+l)+3}=0 \tag{6}$$

for all $\mu=0,1,\ldots,k-1$. Note that (6) can be written as a system of equations:

$$\boldsymbol{A}_{k,l;\alpha}\,\boldsymbol{q}_{k,l;\alpha}=-q^{k}_{k,l;\alpha}\,\boldsymbol{a}_{k,l;\alpha}, \tag{7}$$

where

$$\boldsymbol{A}_{k,l;\alpha}=\begin{bmatrix}\frac{1}{2\alpha l+3} & \frac{1}{2\alpha(1+l)+3} & \cdots & \frac{1}{2\alpha(k-1+l)+3}\\ \frac{1}{2\alpha(1+l)+3} & \frac{1}{2\alpha(2+l)+3} & \cdots & \frac{1}{2\alpha(k+l)+3}\\ \vdots & \vdots & & \vdots\\ \frac{1}{2\alpha(k-1+l)+3} & \frac{1}{2\alpha(k+l)+3} & \cdots & \frac{1}{2\alpha(2k-2+l)+3}\end{bmatrix},\boldsymbol{q}_{k,l;\alpha}=\begin{bmatrix}q^{0}_{k,l;\alpha}\\ q^{1}_{k,l;\alpha}\\ \vdots\\ q^{k-1}_{k,l;\alpha}\end{bmatrix},\boldsymbol{a}_{k,l;\alpha}=\begin{bmatrix}\frac{1}{2\alpha(k+l)+3}\\ \frac{1}{2\alpha(k+1+l)+3}\\ \vdots\\ \frac{1}{2\alpha(2k-1+l)+3}\end{bmatrix}.$$

Finally, with $k'=k$, (5) and (6) give $3q^{k}_{k,l;\alpha}\sum_{\nu=0}^{k}q^{\nu}_{k,l;\alpha}\frac{1}{2\alpha(\nu+k+l)+3}=1$, which is equivalent to:

$$q^{k}_{k,l;\alpha}\left(\boldsymbol{a}^{T}_{k,l;\alpha}\,\boldsymbol{q}_{k,l;\alpha}+\frac{q^{k}_{k,l;\alpha}}{2\alpha(2k+l)+3}\right)=\frac{1}{3}. \tag{8}$$

Combining (7) and (8), we obtain the quadratic equation

$$-(q^{k}_{k,l;\alpha})^{2}\boldsymbol{a}^{T}_{k,l;\alpha}\,\boldsymbol{A}^{-1}_{k,l;\alpha}\,\boldsymbol{a}_{k,l;\alpha}+\frac{(q^{k}_{k,l;\alpha})^{2}}{2\alpha(2k+l)+3}=\frac{1}{3},$$

whose solution can be given by:

$$q^{k}_{k,l;\alpha}=\frac{1}{\sqrt{3\left(\frac{1}{2\alpha(2k+l)+3}-\boldsymbol{a}^{T}_{k,l;\alpha}\,\boldsymbol{A}^{-1}_{k,l;\alpha}\,\boldsymbol{a}_{k,l;\alpha}\right)}}.$$

The remaining coefficients $q^{\nu}_{k,l;\alpha}$ with $\nu=0,\ldots,k-1$ can then be determined by solving the system in (7). It can be proved by induction that these coefficients have the following explicit form:

$$q^{\nu}_{k,l;\alpha} = \frac{(-1)^{k+\nu}}{k!\,(2\alpha)^k}\sqrt{\frac{2\alpha l + 4\alpha k + 3}{3}}\binom{k}{\nu}(2\alpha l + 2\alpha\nu + 3)^{(k)}, \tag{9}$$

where the notation $x^{(k)}$ is the rising factorial (Pochhammer function) defined by $x^{(0)} = 1$ and $x^{(k)} = x(x+1)\dots(x+k-1)$ if $k \in \mathbb{Z}^+$.

Note that when $\alpha = 1$, our work above yields the formulas in [12]. Thus, our new method extends the radial polynomials and, hence, the Zernike functions to allow for fractional exponents. The fractional radial functions in (1) with the newly determined coefficients, $q^{\nu}_{k,l;\alpha}$, now satisfy the orthogonality condition in (4). Combining this result with (3), we conclude:

$$\frac{3}{4\pi}\int_0^1\int_0^{2\pi}\int_0^{\pi} Z^m_{n,l;\alpha}(r,\theta,\phi)\,\overline{Z^{m'}_{n',l';\alpha}(r,\theta,\phi)}\,r^2\sin\theta\,d\theta\,d\phi\,dr = \delta_{nn'}\,\delta_{ll'}\,\delta^{mm'}. \tag{10}$$

The 3D fractional Zernike functions in (2) can be written in Cartesian coordinates as follows:

$$Z^m_{n,l;\alpha}(x,y,z) = S_{k,l;\alpha}(r)\,e^m_l(x,y,z), \tag{11}$$

where $n-l$ is even, $k = (n-l)/2$, and

$$S_{k,l;\alpha}(r) = \tfrac{R_{n,l;\alpha}(r)}{r^l} = r^{(\alpha-1)l}\sum_{\nu=0}^{k} q^{\nu}_{k,l;\alpha} r^{2\alpha\nu} \tag{12}$$

with $r = \sqrt{x^2+y^2+z^2}$. Here, e^m_l denotes the harmonic functions that can be written in Cartesian coordinates as follows:

$$e^m_l(x,y,z) = \sum_{a+b+c\le n} \xi^{a,b,c}_{l,m} x^a y^b z^c, \tag{13}$$

with the coefficients $\xi^{a,b,c}_{l,m}$ explicitly defined in [13]. Then, using the standard convention for converting spherical coordinates to Cartesian coordinates, (10) is now equivalent to

$$\frac{3}{4\pi}\iiint_{B^3} Z^m_{n,l;\alpha}(x,y,z)\,\overline{Z^{m'}_{n',l';\alpha}(x,y,z)}\,dz\,dy\,dx = \delta_{nn'}\,\delta_{ll'}\,\delta^{mm'}, \tag{14}$$

where $B^3 = \{(x,y,z) \in \mathbb{R}^3 : x^2+y^2+z^2 \le 1\}$ is the unit ball. Both (10) and (14) prove that the new fractional Zernike functions are orthogonal in B^3.

3. 3D Fractional-Order Zernike Moments and Shape Descriptors

Let $f(x,y,z)$ be a function defined on the unit ball B^3. 3D fractional-order Zernike moments (3DFZMs) of $f(x,y,z)$ of order αn are defined by:

$$\Omega^m_{n,l;\alpha} = \frac{3}{4\pi}\iiint_{B^3} f(x,y,z)\,\overline{Z^m_{n,l;\alpha}(x,y,z)}\,dz\,dy\,dx, \tag{15}$$

where $n \in \mathbb{N}$, $0 \le l \le n$, $-l \le m \le l$, and $n-l$ is even. For 3D shapes defined by a function $f(x,y,z)$ on a discrete grid in B^3, these moments can be represented by:

$$\Omega^m_{n,l;\alpha} = \frac{3}{4\pi}\sum_x\sum_y\sum_z f(x,y,z)\,\overline{Z^m_{n,l;\alpha}(x,y,z)} \tag{16}$$

with $x^2+y^2+z^2 \le 1$. Note that the 3DFZMs, $\Omega^m_{n,l;\alpha}$, are not invariant under rotations. Also, note that the new parameter α in the definition of 3D fractional Zernike functions

only applies to the radial function $R_{n,l;\alpha}(r)$, and the spherical harmonic function $Y_l^m(\theta,\phi)$ remains intact. Thus, similarly to [13], we can form a $(2l+1)-$dimensional vector

$$\Omega_{n,l;\alpha} = (\Omega_{n,l;\alpha}^{-l}, \Omega_{n,l;\alpha}^{-l+1}, \ldots, \Omega_{n,l;\alpha}^{l-1}, \Omega_{n,l;\alpha}^{l})^T$$

and take its Euclidean norm to define the rotationally invariant 3D fractional-order Zernike descriptors (3DFZDs), $F_{n,l;\alpha}$, of order αn as follows:

$$F_{n,l;\alpha} = \sqrt{\sum_{m=-l}^{l} \left|\Omega_{n,l;\alpha}^{m}\right|^2}. \tag{17}$$

The 3D fractional Zernike functions form an orthogonal basis according to (14). They are constructed via separation of variables into a fractional radial component $R_{n,l;\alpha}(r)$ and the standard spherical harmonics $Y_l^m(\theta,\phi)$. Because the spherical harmonics are known to form a complete orthonormal basis for the angular space $L^2(S^2)$, the completeness of the entire 3D fractional Zernike system for the Hilbert space $L^2(B^3)$ reduces strictly to proving the completeness of the fractional radial functions $R_{n,l;\alpha}(r)$ within the weighted Hilbert space $L_w^2([0,1])$ with the weight function $w(r) = r^2$.

The fractional radial functions in (1) can be rewritten as follows:

$$R_{n,l;\alpha}(r) = \sum_{\nu=0}^{k} q_{k,l;\alpha}^{\nu}\, r^{\alpha(2\nu+l)}, \tag{18}$$

where $n, l \in \mathbb{N}$, $l \le n$, and $n-l$ is even. When $\nu \in \{0,1,2,\ldots,k\}$, we have $\mu = 2\nu + l \in \{l, 2+l, 4+l, \ldots, n\}$. As n and l run through all natural numbers such that $l \le n$ and $n-l$ is even, the number μ spans the entire set of natural numbers.

Consequently, each radial function $R_{n,l;\alpha}(r)$ represents a finite linear combination of $1, r^{\lambda_1}, r^{\lambda_2}, r^{\lambda_3}, \ldots$, where $\{\lambda_\mu\} = \{\alpha\mu\}_{\mu=1}^{\infty}$ is an increasing sequence of positive real numbers. Because $\alpha > 0$ and the standard harmonic series diverges to infinity, the series is as follows:

$$\sum_{\mu=1}^{\infty} \frac{1}{\lambda_\mu} = \frac{1}{\alpha} \sum_{\mu=1}^{\infty} \frac{1}{\mu} = +\infty.$$

Hence, by the Müntz–Szász Theorem [21], the linear span of the sequence 1, r^{λ_1}, r^{λ_2}, r^{λ_3}, … is dense in the continuous function space $\mathcal{C}([0,1])$—and by extension, dense in the weighted space $L_w^2([0,1])$. Since the coefficients $q_{k,l;\alpha}^{\nu}$ are explicitly chosen to satisfy the orthogonality condition under the measure $r^2\,dr$ via re-orthogonalization, the sequence forms a complete orthogonal basis for the radial Hilbert space $L_w^2([0,1])$.

Therefore, the 3D fractional Zernike functions $Z_{n,l;\alpha}^{m}$ form a complete orthogonal basis on the unit ball B^3, and hence any square-integrable function $f(x,y,z)$ in B^3 can be expanded as follows:

$$f(x,y,z) = \sum_{n=0}^{\infty} \sum_{l=0}^{n} \sum_{m=-l}^{l} \Omega_{n,l;\alpha}^{m} Z_{n,l;\alpha}^{m}(x,y,z), \tag{19}$$

where $n-l$ is an even number. In practice, however, we can only compute truncated approximations by using a finite number of 3DFZMs for a given reconstruction order N as follows:

$$f_N(x,y,z) = \sum_{n=0}^{N} \sum_{l=0}^{n} \sum_{m=-l}^{l} \Omega_{n,l;\alpha}^{m} Z_{n,l;\alpha}^{m}(x,y,z). \tag{20}$$

4. Computation of 3D Fractional-Order Zernike Moments

The 3D fractional Zernike functions are given in Cartesian coordinates in (11). Note that the function $S_{k,l;\alpha}$ in (12) depends on r and, because of fractional exponents, it cannot be written in terms of the monomials $x^a y^b z^c$. Thus, direct computation of 3DFZMs using (16) requires element-wise multiplications and exponentiations of non-separable 3D functions and can be quite time-consuming, especially when the grid size for the (x, y, z)-variables is large. For this reason, we introduce a recurrence relation framework for efficient computation of 3DFZMs. Note that:

$$S_{k,l;\alpha}(r) = \sum_{\nu=0}^{k} T^{\nu}_{k,l;\alpha}(r),$$

where $T^{\nu}_{k,l;\alpha}(r) = q^{\nu}_{k,l;\alpha} r^{2\alpha\nu+(\alpha-1)l}$. Using (9), we derive a three-way recurrence relation to compute $T^{\nu}_{k,l;\alpha}(r)$, and hence $S^{\nu}_{k,l;\alpha}(r)$, as follows:

- $T^{0}_{0,0;\alpha}(r) = 1$
- $T^{0}_{0,l;\alpha}(r) = \sqrt{\dfrac{2\alpha l + 3}{2\alpha(l-1)+3}}\, r^{\alpha-1}\, T^{0}_{0,l-1;\alpha}(r)$ for $1 \le l \le n$
- $T^{0}_{k,l;\alpha}(r) = -\sqrt{\dfrac{2\alpha l + 4\alpha k + 3}{2\alpha l + 4\alpha(k-1)+3}}\, \dfrac{2\alpha l + 2\alpha(k-1)+3}{2\alpha k}\, T^{0}_{k-1,l;\alpha}(r)$ for $1 \le k \le \lfloor \frac{n-l}{2} \rfloor$
- $T^{\nu}_{k,l;\alpha}(r) = -\dfrac{(k-\nu+1)(2\alpha l + 2\alpha(\nu+k-1)+3)}{\nu(2\alpha l + 2\alpha(\nu-1)+3)}\, r^{2\alpha}\, T^{\nu-1}_{k,l;\alpha}(r)$ for $1 \le \nu \le k$

Once we obtain an $S_{k,l;\alpha}$ function, we multiply it by $f(x, y, z)$, and then we handle the rest of the computations efficiently by reordering the sums in (16) and eliminating one dimension at a time, namely

$$\Omega^{m}_{n,l;\alpha} = \frac{3}{4\pi} \sum_{a+b+c \le n} \xi^{a,b,c}_{l,m} \sum_{x} x^a \sum_{y} y^b \sum_{z} z^c f(x, y, z) S_{k,l;\alpha}(r). \tag{21}$$

Note that the coefficients $\xi^{a,b,c}_{l,m}$ are from (13) and do not depend on the input shape function $f(x, y, z)$. Thus, we precompute them and store the nonzero ones in a list to perform the final summation in (21) more quickly. Also, the 3D fractional Zernike functions in (11) and 3DFZMs in (21) have three indices, but we convert them to carry a single index using the algorithm given in [22].

5. Results

In this section, we compare the retrieval performance of 3DFZMs with that of traditional 3D Zernike moments (3DZMs) by applying them to the reconstruction of test shapes. We download the triangulated surface models for these shapes from the benchmark dataset in [23] and map them onto a 3D cubic grid of size V^3. Each voxel (a cube defined by the grid) is assigned 1 or 0: 1 for a surface voxel that is located closer than a $\sqrt{3}/2$ grid interval to any triangle defining the surface and 0 otherwise. This grid of 1s and 0s represents the 3D function $f(x, y, z)$ for the shape.

Here, we define the discrete 3D grid (x, y, z) with

$$x, y, z \in \{-1 + 2i/(V-1) \,|\, i = 0, 1, \ldots, V-1\}$$

so that the voxelized shape is mapped to a grid of size V^3 with $-1 \le x, y, z \le 1$ inscribing the unit ball. Next, we normalize the 3D shape by dividing the density function $f(x, y, z)$ by the zero-order geometric moment M_{000} so that its total mass equals 1. Then, the

center of mass of the shape can be found using the first-order geometric moments, namely $(\bar{x}, \bar{y}, \bar{z}) = (M_{100}, M_{010}, M_{001})$. We redefine the 3D grid as $(x - \bar{x}, y - \bar{y}, z - \bar{z})$ so that the center of mass of the shape is at the origin. Finally, we compute the distance d from the origin to the furthest voxel of the shape for which $f(x, y, z)$ is not zero. We scale the axes with d, namely using the 3D grid $(\frac{x-\bar{x}}{d}, \frac{y-\bar{y}}{d}, \frac{z-\bar{z}}{d})$; so, the shape is fully inscribed inside the unit ball. The 3DFZMs for these voxelized shapes are computed in the final grid using (21) and then used in (20) to obtain a truncated reconstruction $f_N(x, y, z)$ of order N. To compare the reconstruction to the original shape, we use the relative mean squared error:

$$\text{RelMSE}(f, f_N) = \frac{\sum\limits_{x,y,z} [f(x, y, z) - f_N(x, y, z)]^2}{\sum\limits_{x,y,z} [f(x, y, z)]^2}. \tag{22}$$

To ensure RelMSE behaves consistently, we normalize the reconstruction so its intensity scale aligns with the original shape. Thus, we fit the reconstruction to the scale of the original shape by using ordinary least squares and finding a scaling factor a and a shift b that minimizes the error between the reconstruction and original shape:

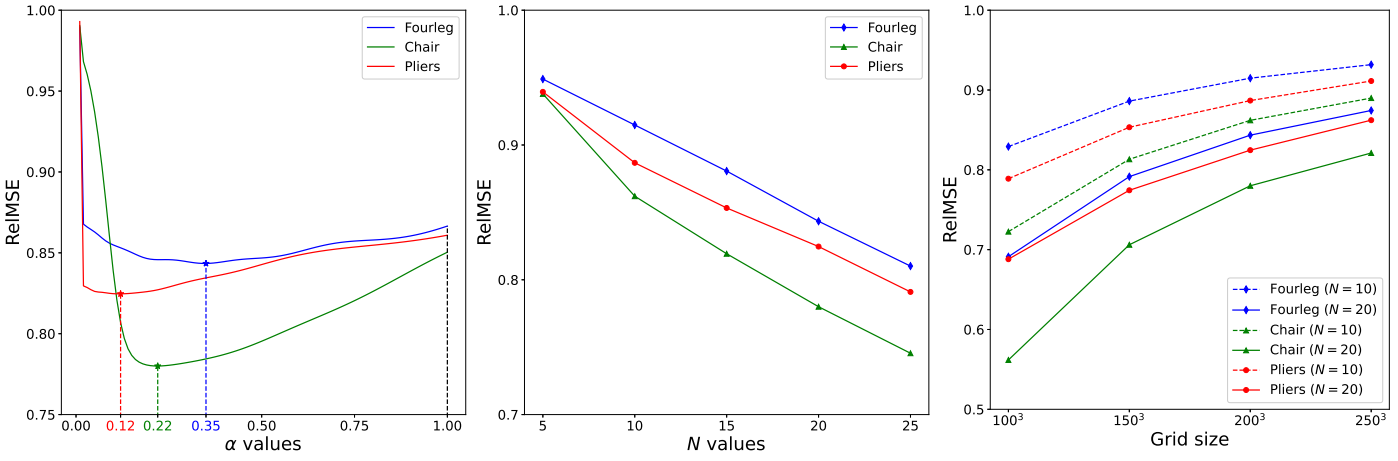
$$\hat{f}_N(x, y, z) = a \cdot f_N(x, y, z) + b, \tag{23}$$

where $a = \text{Cov}(f, f_N)/\text{Var}(f_N)$ and $b = \text{Mean}(f) - a \cdot \text{Mean}(f_N)$. We use $\text{RelMSE}(f, \hat{f}_N)$ as the reconstruction error, namely the relative mean squared error between the original shape and the calibrated reconstruction. The resulting RelMSE purely measures structural shape errors, ensuring it decreases consistently with higher reconstruction orders.

Figure 1 summarizes the key modeling aspects of the 3D fractional-order Zernike framework and motivates the introduction of the fractional radial exponent α. The curves in Figure 1a illustrate this modeling behavior by showing how reconstruction error varies with α for three representative shapes. We set the grid size to 200^3 and the reconstruction order to $N = 20$ and vary α from 0 to 1 with increments of 0.01. We select a model shape from three classes (fourleg, chair, and pliers) in the benchmark dataset and compute its reconstruction and RelMSE corresponding to each α value. Each shape exhibits a clear minimum, indicating that the optimal fractional exponent is shape-dependent and that fractional radial scaling reduces reconstruction error compared with the classical case $\alpha = 1$.

Varying α changes the density of radial nodes within the unit ball. When $\alpha < 1$, the function r^α compresses the radial metric near the boundary ($r \approx 1$) and expands it near the origin ($r \approx 0$), effectively increasing the spatial resolution for central structural features. Because discrete grids introduce partial-volume and staircase artifacts along surfaces, the smooth, non-integer decay of fractional radial functions acts as an implicit smoothing filter, making them less sensitive to voxelization grid alignment than sharp integer polynomials.

Figure 1b,c further characterizes the model by examining how reconstruction accuracy depends on the reconstruction order N and the voxel grid size. Using the optimal α values identified in Figure 1a, RelMSE decreases monotonically as N increases from 5 to 25, confirming that higher-order fractional moments encode progressively finer geometric detail. Conversely, for a fixed reconstruction order, RelMSE increases as the grid size increases from 100^3 to 250^3, indicating that finer voxel grids contain more geometric variation than a fixed number of moments can represent. Among the tested resolutions, the 200^3 grid provides the most balanced performance: it is fine enough to capture essential geometric structure while keeping RelMSE lower than the 250^3 grid for the same reconstruction order N. Thus, 200^3 represents an effective practical resolution where reconstruction accuracy remains competitive without incurring the higher computational cost and increased relative error observed at 250^3.



(**a**) (**b**) (**c**)

Figure 1. Parameter tuning for accurate and efficient reconstruction using 3DFZMs. (**a**) Shape-specific α-optimization. RelMSE of reconstructions obtained with reconstruction order $N = 20$ for three benchmark shapes: Fourleg (Model 395), Chair (Model 105), and Pliers (Model 208). For each shape, the fractional parameter $\alpha \in (0, 1]$ yielding the lowest RelMSE is indicated. (**b**) Reconstruction order (N) vs. RelMSE for these shapes. Reconstructions are generated (with the optimal α values) using $N = 5, 10, \ldots, 25$ and the grid size of 200^3. (**c**) Grid size vs. RelMSE for these shapes. Reconstructions are generated using $N = 10, 20$ and grid sizes of 100^3, 150^3, 200^3, and 250^3.

Having tuned the fractional parameter α, the reconstruction order N, and the grid size, we now compare 3DFZMs with the classical 3DZMs. Figure 2 presents reconstructions of the aforementioned three shapes with reconstruction orders of $N = 10$, 15, and 20 using 3DFZMs and 3DZMs. We compare each reconstruction to the original shape using RelMSE. The reconstructions from 3DFZMs exhibit lower RelMSE (i.e., greater resemblance to the original shapes) and capture more structural details than those from 3DZMs in all the instances shown in Figure 2. The results also suggest that shapes with thin, elongated, or branching protrusions benefit from smaller fractional powers, which allow lower-order functions to capture rapid high-frequency geometric transitions without requiring computationally expensive higher orders ($N > 20$).

Because the optimal α can be different for each shape, it will be more beneficial to find a global α value for which the retrieval performance is the highest for the entire benchmark dataset. We will use 3DFZDs to find the optimal α since their computations and comparisons are far more efficient than the computations and voxel-to-voxel comparisons of reconstructions. To evaluate the discriminative performance of 3DFZDs, we leverage their rotational invariance. Using (17), we form the feature vector of fractional-order descriptors:

$$\vec{F}_\alpha = \langle F_{n,l;\alpha} \,:\, 0 \le n \le N,\ 0 \le l \le n,\ n - l \text{ is even} \rangle. \tag{24}$$

We set $N = 20$, yielding 121 such descriptors in the feature vector. Note that when $\alpha = 1$, they become the 3D Zernike descriptors (3DZDs) in [13]. We employ the squared Euclidean distance for the similarity measure between two feature vectors (hence two shapes), namely

$$d\left(\vec{F}_\alpha^{\,Q}, \vec{F}_\alpha^{\,T}\right) = \sum_{n=0}^{20} \sum_{l=0}^{n} \left(F_{n,l;\alpha}^{Q} - F_{n,l;\alpha}^{T}\right)^2, \tag{25}$$

where $n - l$ is even, and Q and T stand for a query and a target shape to be compared.

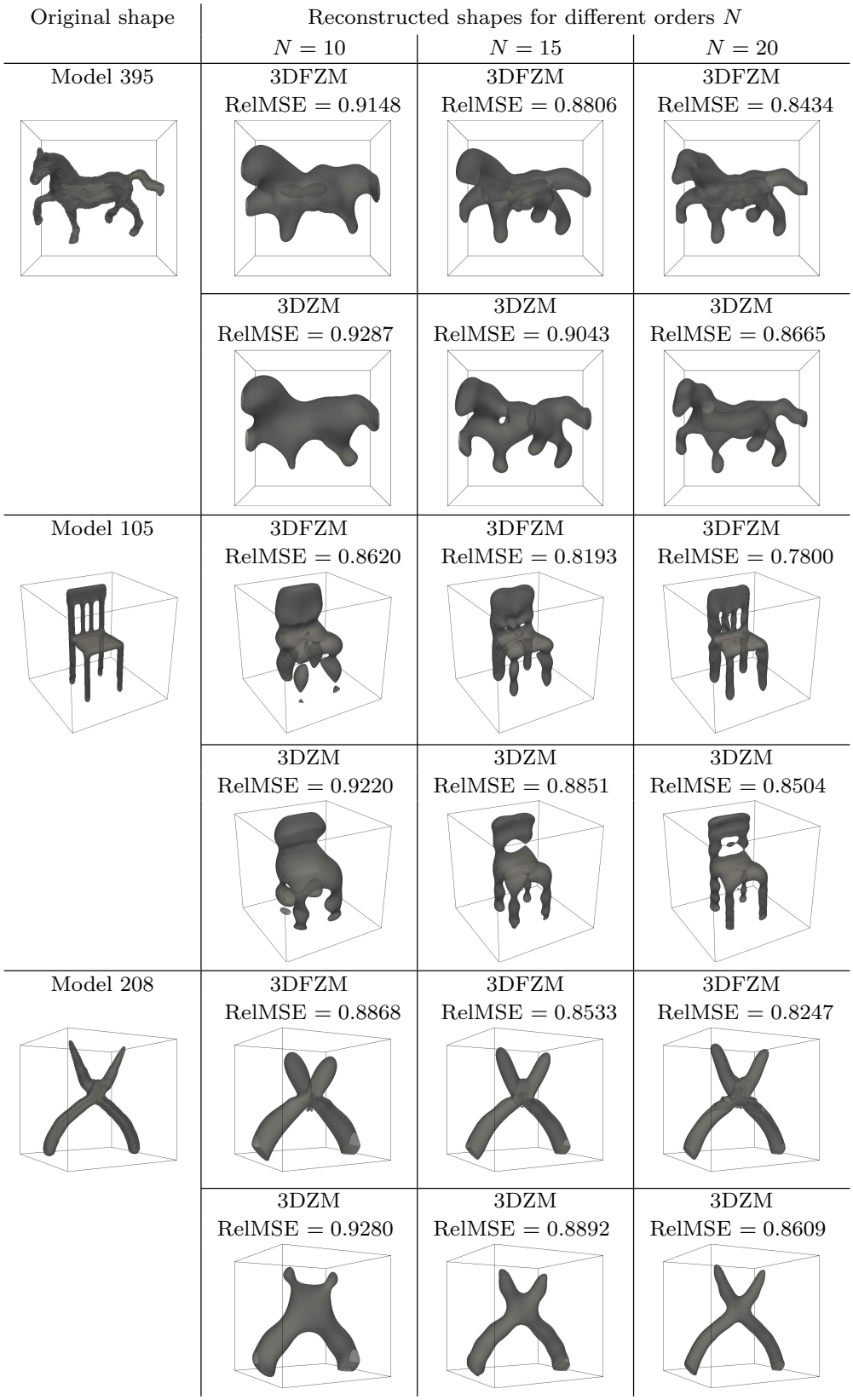


Figure 2. Examples of 3D shapes and their reconstructions using (20) with orders $N = 10$, 15, and 20. Reconstructions are demonstrated for 3DFZMs (with the optimal α from Figure 1a) and traditional 3DZMs. The grid size for each shape is 200^3. To ensure an unbiased comparison across reconstructions, we adaptively set each volume's isosurface threshold to the 5th percentile of intensity values sampled strictly within a 2-voxel dilated neighborhood of the original shape layout. Above each reconstruction, we show the corresponding RelMSE.

To find the α value that yields the highest recognition accuracy, we download the entire benchmark dataset in [23], comprising 380 shapes across 19 classes (e.g., human, cup, ant), with 20 shapes per class. We map each shape to the unit ball in a grid as described above. We compute the 3DFZDs of all shapes for α values ranging from 0 to 1 in increments of 0.01 and store the descriptors corresponding to each α value. Then, for each α, we scale the precomputed data by using MinMaxScaler so that descriptors with larger numerical ranges do not dominate the distance-based similarity metric. We then perform a nearest-neighbor search by querying each of the 380 shapes and comparing it to the remaining 379 shapes using the distance metric in (25). If the shape with the minimum distance to the query (the nearest neighbor) is of the same class as the query, we label the query as "correctly classified." Next, for each α, we compute recognition accuracy as the percentage of query shapes correctly classified. The results are summarized in Figure 3, in which accuracy peaks at $\alpha = 0.14$, 0.20, and 0.38. Each of these α values yields the highest average recognition accuracy of 87.6%, outperforming classical 3D Zernike descriptors ($\alpha = 1$), which achieve 83.9%.

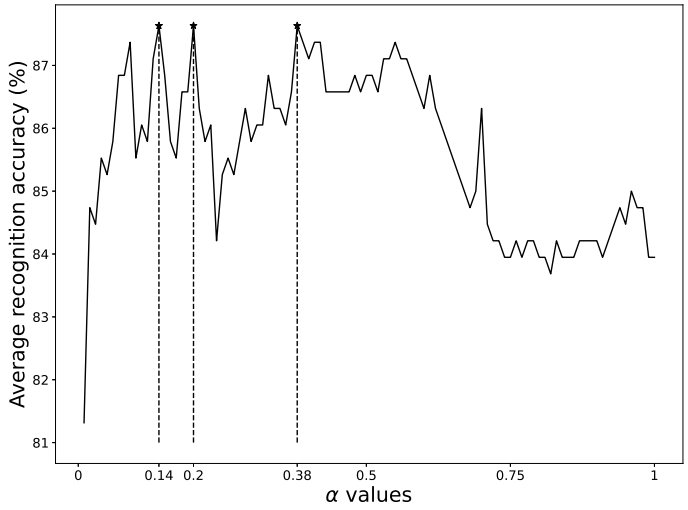


Figure 3. Average recognition accuracy across the 380 shape benchmark dataset for 3DFZDs as the fractional parameter α varies from 0 to 1. Accuracy is computed using nearest-neighbor classification with squared Euclidean distance between descriptor vectors. The highest average accuracy, 87.6%, occurs at $\alpha = 0.14$, 0.2, and 0.38.

The fluctuations in Figure 3 arise from the heterogeneity of the benchmark dataset, which contains 19 shape classes with widely varying geometric complexity. Each class exhibits a different optimal fractional exponent α; so, averaging recognition accuracy across all classes produces a non-monotonic global curve. The small magnitude of accuracy changes reflects the fact that class-specific improvements partially cancel each other when aggregated. Accuracy decreases for $\alpha > 0.38$ because larger fractional exponents amplify high-frequency radial components, which increases sensitivity to voxelization noise and disproportionately affects thin or elongated shapes. These class-dependent behaviors collectively explain the observed fluctuations and the mild degradation beyond $\alpha \approx 0.38$.

Since Figure 3 recommends three global α values, we select the optimal one by examining recognition accuracy for each class and present the results in Table 1. Note that the 3DFZDs with $\alpha = 0.14$ and 0.38 outperform 3DZDs in 9 classes (out of 19), whereas it remains at 8 for 3DFZDs with $\alpha = 0.20$. To break the tie between $\alpha = 0.14$ and 0.38, we compare the number of times 3DFZDs are outperformed by 3DZDs: 2 classes (Human and Bird) for $\alpha = 0.38$, but 4 classes (Octopus, Bird, Armadillo, and Bearing) for $\alpha = 0.14$.

Thus, $\alpha = 0.38$ is the unique global parameter that maximizes both overall and per-class recognition accuracy.

Table 1. Classification accuracy comparison of 3DFZDs and 3DZDs on the benchmark dataset [23].

Descriptor	Human	Cup	Glasses	Airplane	Ant
3DFZDs ($\alpha = 0.14$)	18	17	19	20	20
3DFZDs ($\alpha = 0.20$)	18	16	19	20	19
3DFZDs ($\alpha = 0.38$)	17	16	19	20	20
3DZDs	18	16	19	17	18
Descriptor	Chair	Octopus	Table	Teddy	Hand
3DFZDs ($\alpha = 0.14$)	20	13	16	20	18
3DFZDs ($\alpha = 0.20$)	20	14	16	20	18
3DFZDs ($\alpha = 0.38$)	20	14	16	20	19
3DZDs	19	14	12	20	17
Descriptor	Plier	Fish	Bird	Armadillo	Bust
3DFZDs ($\alpha = 0.14$)	20	20	12	18	18
3DFZDs ($\alpha = 0.20$)	20	20	13	19	18
3DFZDs ($\alpha = 0.38$)	20	20	11	19	18
3DZDs	20	20	16	19	18
Descriptor	Mech	Bearing	Vase	Fourleg	Average
3DFZDs ($\alpha = 0.14$)	19	15	13	17	87.6%
3DFZDs ($\alpha = 0.20$)	18	15	12	18	87.6%
3DFZDs ($\alpha = 0.38$)	18	17	12	17	87.6%
3DZDs	16	16	8	16	83.9%

Our method boosts discriminative power in classes with subtle geometric variations, such as Table (16 vs. 12) and Vase (12 vs. 8). In categories like Bird (11 vs. 16), the global optimal parameter $\alpha = 0.38$ performs below the classical baseline. This suggests that a uniform global α across highly heterogeneous datasets can over-smooth specific volumetric distributions. This highlights the opportunity for future work in class-adaptive fractional descriptors. Overall, the results in this section show that extending 3D Zernike moments and descriptors to fractional ones improves their discriminative power and accuracy in representing 3D shapes.

6. Comparison to a Recent Method

Efficient computation is a critical requirement in 3D pattern recognition and volumetric image analysis, where descriptor extraction must remain tractable for large datasets. CPU time therefore serves as an essential quantitative metric for evaluating the practical feasibility of any moment-based method. In this work, we implement all numerical experiments in Python 3.14 (64-bit) on a Windows laptop with an Intel(R) Core(TM) Ultra 5 225U (1.50 GHz) and 16 GB of memory. To contextualize the computational efficiency of 3DFZDs relative to existing invariant descriptors, we compare them with the 3D orthogonal moment invariants (3dOMIs) of Hjouji et al. [24].

Table 2 demonstrates a performance comparison between the proposed 3DFZDs for $\alpha = 0.38$ and the non-separable 3dOMIs. For each descriptor order, the table reports the number of invariants, nearest-neighbor classification accuracy on the 380 shape benchmark dataset (with each shape mapped to a 128^3 grid, as in [24]), and average CPU time per shape. Across all descriptor orders, the 3DFZDs consistently outperform the non-separable 3dOMIs. For $N = 7$, 9, and 11, 3DFZDs improve nearest neighbor classification accuracy by 29.2, 26.9, and 22.7 percentage points, respectively, compared with 3dOMIs.

In addition to higher discriminative power, 3DFZDs require substantially less computation time. Average CPU time per shape is reduced by factors of 2× ($N = 7$), 6.6× ($N = 9$), and 16.3× ($N = 11$), reflecting the efficiency gained from the three-way recurrence relation that eliminates repeated evaluation of fractional radial functions. Overall, Table 2 highlights that 3DFZDs not only deliver markedly superior classification accuracy but also achieve order-of-magnitude reductions in runtime, making them more practical for large-scale 3D shape retrieval.

Table 2. Classification accuracy and runtime comparison of 3DFZDs ($\alpha = 0.38$) vs. 3dOMIs [24].

Descriptor	Order	Number of Invariants	Classification Accuracy	Average CPU Time (s)
3DFZD	$N = 7$	20	82.9%	0.261464
3dOMI	$p = 20$	20	53.7%	0.531266
3DFZD	$N = 9$	30	84.5%	0.355616
3dOMI	$p = 30$	30	57.6%	2.364938
3DFZD	$N = 11$	42	85.3%	0.540518
3dOMI	$p = 42$	42	62.6%	8.782550

7. Conclusions

In this paper, we introduced a novel 3D fractional-order Zernike moment framework by extending the radial basis of classical 3D Zernike polynomials to fractional exponents. We proved that the new fractional Zernike functions form a complete orthogonal basis for the square-integrable functions in the unit ball. To make the framework practical for high-order shape analysis, we derived a three-way recurrence relation that drastically reduces computational cost, eliminating repeated evaluation of fractional radial functions. Furthermore, we constructed rotationally invariant 3D fractional-order Zernike descriptors (3DFZDs) for orientation-independent shape retrieval. Experimental evaluations on a shape benchmark dataset demonstrate that 3DFZDs achieve higher recognition accuracy than classical integer-order 3D Zernike descriptors (3DZDs) and 3D orthogonal invariant moments (3dOMIs). Future work will focus on principled selection of the fractional parameter, developing adaptive or multi-spectral descriptors, and validation in domains where fine structural detail is critical, such as medical and biological volumetric data.

Author Contributions: Conceptualization, A.S.-S.; methodology, A.S.-S.; software, M.G.A. and A.S.-S.; validation, M.G.A. and A.S.-S.; formal analysis, M.G.A. and A.S.-S.; investigation, A.S.-S.; resources, A.S.-S.; data curation, M.G.A. and A.S.-S.; writing—original draft preparation, M.G.A. and A.S.-S.; writing—review and editing, A.S.-S.; visualization, A.S.-S.; supervision, A.S.-S.; project administration, A.S.-S.; funding acquisition, M.G.A. and A.S.-S. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Directorate for Mathematical and Physical Sciences (MPS) at the National Science Foundation (NSF), grant number 2151679. M.A. also acknowledges financial support from the Summer Scholars Program at Saint Joseph's University.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The 3D benchmark shape dataset used in this study is currently available at https://segeval.cs.princeton.edu.

Acknowledgments: During the preparation of this manuscript/study, the author(s) used Grammarly for the purpose of correcting grammatical or spelling errors. The authors have reviewed and edited the output and take full responsibility for the content of this publication.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

2D	Two-dimensional
3D	Three-dimensional
3DZM	3D Zernike moment
3DFZM	3D fractional-order Zernike moment
3DZD	3D Zernike descriptor
3DFZD	3D fractional-order Zernike descriptor
3dOMI	3D orthogonal invariant moment
RelMSE	Relative mean squared error

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